

# BORDER OPTICS

## A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)

Executive Summary · DOI: 10.5281/zenodo.21759970 · Sakshi D. Maske

### The Question

Asked directly in Parliament whether the Vibrant Villages Programme's impact had ever been assessed, the Ministry of Home Affairs answered without qualification: "No impact assessment has been carried out" (Lok Sabha Unstarred Question No. 508, 3 February 2026). Did India's Rs. 4,800 crore border-village development scheme produce measurable physical development on the ground — and can the answer be trusted, or does it depend on an arbitrary measurement choice?

### The Method

258 individually geocoded villages across five Himalayan border states/UTs (Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh, Ladakh) were tested using Sentinel-2 built-up-area change (NDBI) and VIIRS night-lights, 2021-2025. Every test was deliberately run under two independent satellite compositing windows — full-year and summer-matched — specifically to test whether the methodology choice itself changes the answer, using Wilcoxon signed-rank tests for before/after change and Spearman correlations for budget and border-proximity relationships.

### The Finding

Built-up area change is compositing-window sensitive, not confirmed. A full-year composite shows no significant increase ( $p = 1.000$ ); the identical villages under a summer-matched composite show a highly significant increase ( $p < 0.000001$ ) — traced to snow-cover contamination in the full-year window and monsoon cloud cover eliminating Sikkim's data in the summer window. This instability is reported as the project's central finding, not resolved by preferring one window.

| Test                           | Full-Year Window              | Summer-Matched Window               |
|--------------------------------|-------------------------------|-------------------------------------|
| NDBI change (Wilcoxon)         | $p = 1.000$ — not significant | $p < 0.000001$ — highly significant |
| Night-lights change (Wilcoxon) | $p = 0.050$                   | $p = 0.9999$                        |
| Budget vs. measured change     | ~10x budget gap (AP vs. UK)   | Nearly identical mean change        |

*Night-lights, the more temporally stable proxy, shows a consistent null across both windows — carrying more weight than the window-sensitive NDBI result. Arunachal Pradesh's sanctioned budget (Rs. 2,749.74 crore) is roughly ten times Uttarakhand's (Rs. 270.58 crore), yet the two states' mean measured built-up change is nearly identical — descriptively consistent with budget-independent implementation.*

### Validation & Robustness Checklist

- ✓ Dual compositing-window test (full-year + summer-matched)
- ✓ Two independent metrics tested separately (NDBI + VIIRS)
- ✓ Wilcoxon signed-rank tests (paired before/after)
- ✓ Cross-checked against sanctioned budget (Parliamentary record)
- ✓ Every data gap disclosed (Ladakh, HP, Uttarakhand caveats)
- ! NDBI result flagged as window-sensitive — not confirmed (the project's central finding)

### Honest Limitation

This project's own headline result is an instability disclosure rather than a clean confirmation: Himalayan terrain introduces genuine seasonal measurement artifacts (snow cover in winter months, monsoon cloud cover eliminating usable Sikkim data in summer) that make a single compositing choice unreliable evidence on its own. The border-proximity (H3) correlation is explicitly treated as exploratory given sample size and border-geometry caveats. Data gaps — Ladakh's unresolved villages, Himachal Pradesh's partial sample, Uttarakhand's block ambiguity — are disclosed rather than smoothed over.

### Real-World Relevance

This project is the independent impact assessment Parliament was told does not exist for a Rs. 4,800 crore government scheme spanning 2,967 sanctioned villages. It demonstrates that satellite-based verification of public infrastructure spending is both feasible and necessary — and that when the two most defensible measurement choices produce opposite conclusions, disclosing that instability is more useful to policymakers than a false confidence in either single result.

**GitHub:** [github.com/sakshimaske303-commits/BORDER\\_OPTICS](https://github.com/sakshimaske303-commits/BORDER_OPTICS) | **Live Dashboard:**

[borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app](https://borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app) | **Zenodo DOI:** 10.5281/zenodo.21759970

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